TRAIN DELAY FORECASTING · NYC SUBWAY

Direct Multi-Horizon GATv2 for Train Delay Prediction

Predicting the next three train events at once, so errors stop piling up

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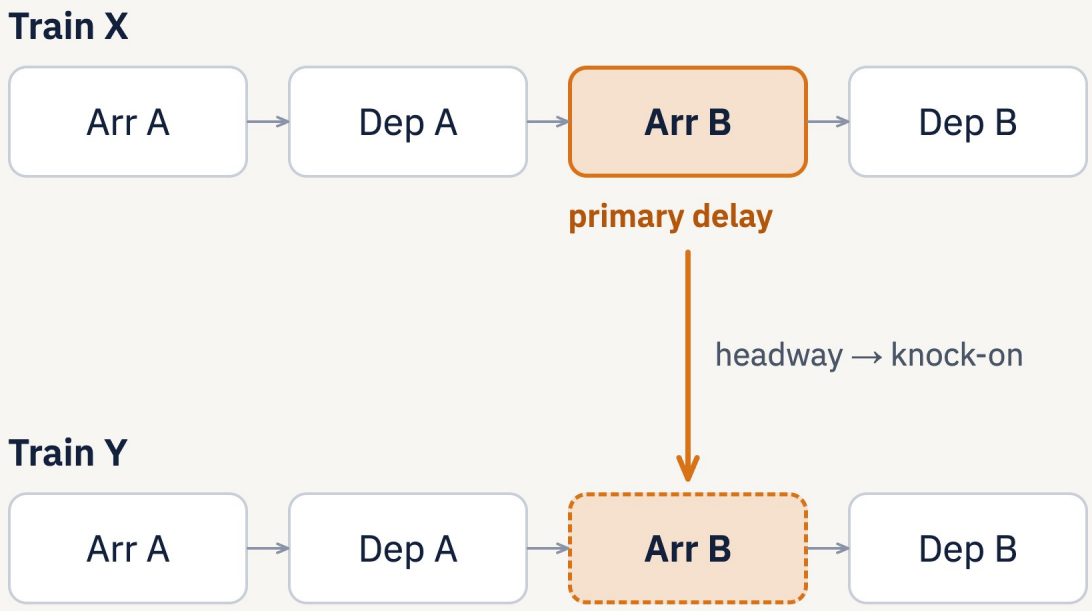
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01 · MOTIVATION

One late train makes others late

Trains share tracks, platforms and headway limits. A **primary delay** on one train spreads to the next train as a **knock-on delay**, and can cascade across a dense network.

Forecasting a few events ahead gives operators time to react.



Delay propagation between two trains, adapted from Nguyen & Li (2025)

01 · THE PROBLEM

Autoregressive rollout lets errors pile up

An autoregressive (AR) model predicts one step at a time and feeds each prediction back in as the input for the next step.

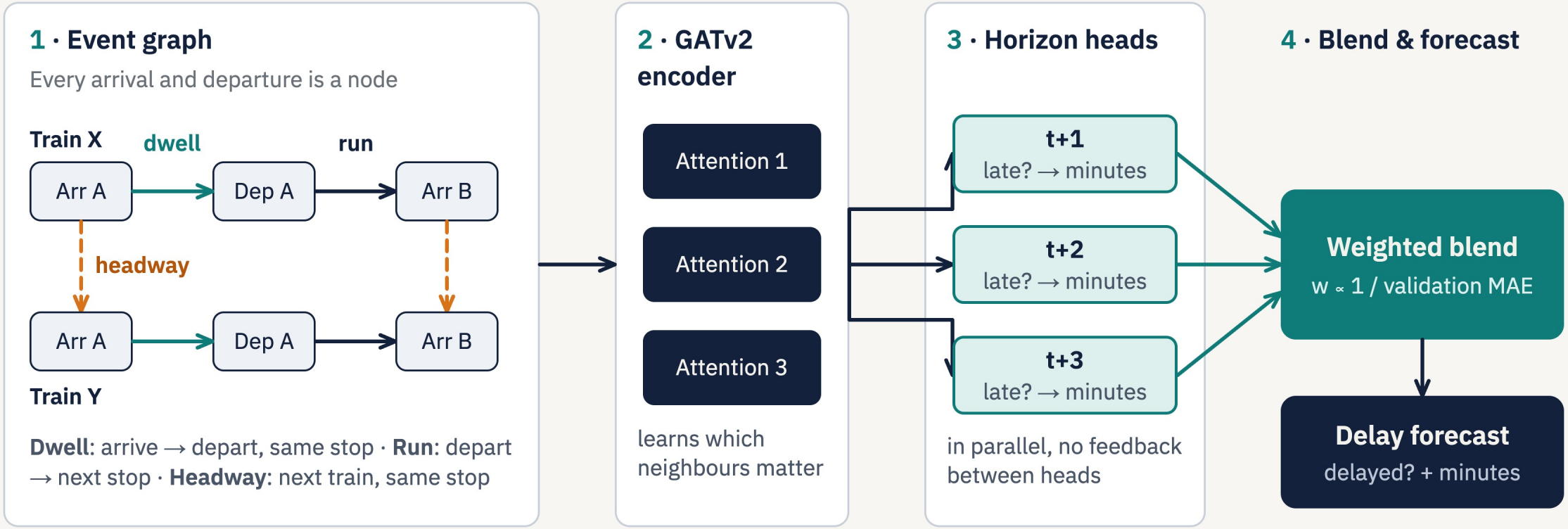


AR mean absolute error by horizon: error grows about 30% from t+1 to t+3.

Research question: can direct multi-horizon decoding reduce error at later horizons?

02 · OUR MODEL

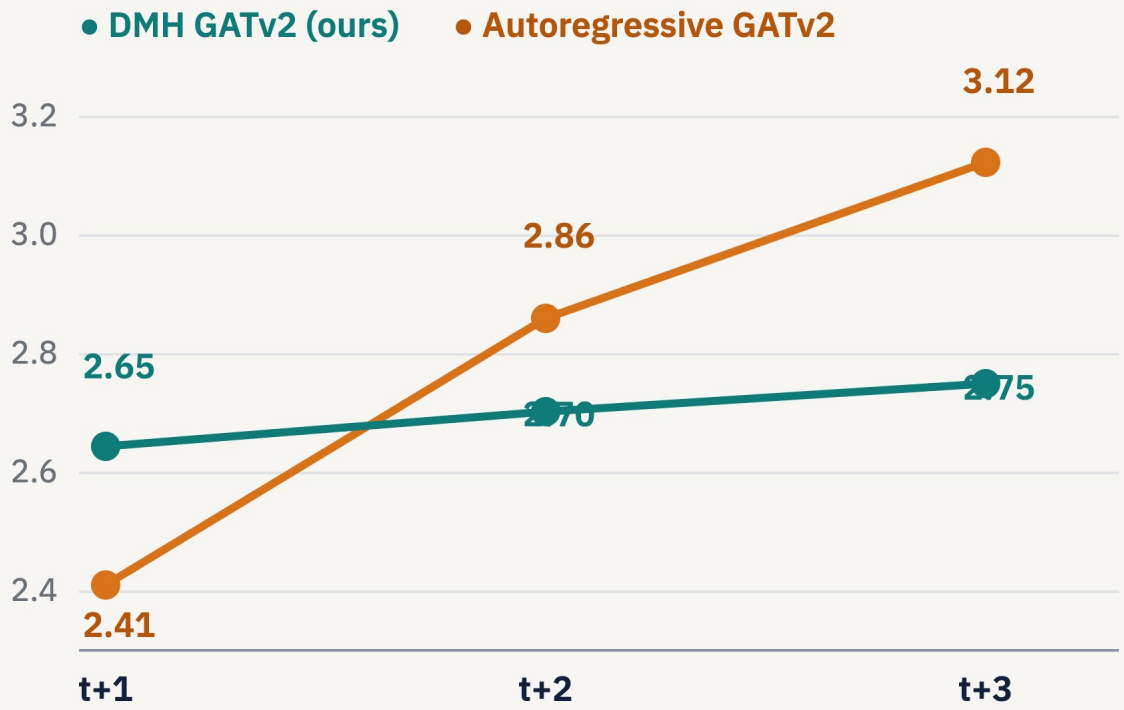
One graph, three horizons at once, no feedback



Each head is a two-stage hurdle: first classify whether the event is late, then regress how many minutes.

03 · RESULTS

DMH error stays nearly flat across horizons



MAE in minutes, lower is better

6.7×

less error growth from t+1 to t+3 (+0.11 vs +0.71 min)

12%

lower MAE at t+3 (2.75 vs 3.12 min)

Trade-off: at t+1 the autoregressive model is slightly better.

03 · OVERALL PERFORMANCE

DMH detects delays best of all models

Model	MAE	RMSE	Accuracy	Precision	Recall	F1
Zero-delay	6.710	13.711	0.517	0.000	0.000	0.000
Persistence	**0.729**	**1.893**	0.929	0.938	0.913	0.926
Autoregressive GATv2	3.124	7.597	0.922	0.938	0.899	0.918
DMH GATv2 + aggregation (ours)	2.560	6.914	**0.946**	**0.949**	**0.938**	**0.944**

Overall validation performance; MAE and RMSE in minutes. **Bold orange** = best value in each column.

+3.9 recall points over autoregressive

93.8% vs 89.9%: DMH misses fewer real delays.

Persistence still wins raw MAE

Consecutive subway delays rarely jump, so "no change" is a strong guess.

04 · CONCLUSION

Direct decoding curbs error accumulation

+4% vs +30%

MAE growth from t+1 to t+3, DMH vs autoregressive

94.4% F1

Best delay detection of all models; persistence still wins raw MAE

Next

Tune the horizon length; test on networks with more sudden disruptions

Thank you

Questions?

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